

1.2 NLP

1.2.1 Text Dataset Selection and Preprocessing

For this part of a project, we selected a publicly available dataset containing Steam reviews for the game *Counter-Strike: Global Offensive (CS:GO)* taken between 31st January 2023 to 1st March 2023. The dataset is available on Kaggle: <https://www.kaggle.com/datasets/noahxl1/csgo-steam-reviews>

It contains 15,640 user reviews, each labelled with a binary variable `voted_up`, where 1 represents a positive recommendation and 0 represents a negative recommendation. The dataset is highly imbalanced, with approximately 89% positive reviews and 11% negative reviews, which makes it suitable for a binary sentiment analysis task. Reviews are typically very short, with an average length of around 5–6 words and approximately 47 characters per review, reflecting the concise nature of many Steam user comments.

Before training the models, several preprocessing steps were applied to improve data quality. First, reviews with missing or empty text were removed. Reviews written mostly in non-Latin, for example Cyrillic or ASCII art were filtered out by computing a Latin character ratio and removing reviews with a ratio below 0.15. The remaining reviews were then cleaned by converting all text to lowercase, removing any URLs, HTML tags, punctuation, and non-alphabetic characters using regular expressions. The text was tokenized, and common English stop words were removed using the NLTK stop word list to reduce noise and improve feature quality. Reviews that became empty after cleaning were also discarded. The cleaned text was stored as a new feature and the sentiment label was converted into a binary target variable for model training.

For feature representation, pre-trained word embeddings from Global Vectors for Word Representation (GloVe) were used. Specifically, the GloVe 6B dataset with 100-dimensional vectors was downloaded from the Stanford NLP repository. Each word in the vocabulary was mapped to its corresponding embedding vector to create an embedding matrix, which was then used to initialise the embedding layer in the neural network models. Using pre-trained embeddings allows the model to leverage semantic relationships learned from large text corpora, improving generalization and enabling the model to capture contextual similarities between words.

1.2.2 Deep Learning Model Implementation

Following the preprocessing and feature representation, deep learning models were designed and trained to perform sentiment classification on the CS:GO Steam reviews dataset. The objective of the models is to predict whether a review expresses a positive or negative sentiment, based on the binary label. Text reviews represent sequential data, neural network architectures capable of modelling word order and contextual relationships were used.

To prepare the input for the neural networks, a custom vocabulary was constructed from the cleaned review corpus. The vocabulary size was limited to the 20,000 most frequent words, ensuring that the models focus on the informative tokens while keeping computational complexity manageable. Each review was converted into a sequence of integer indices representing the words in the vocabulary. Because neural networks require inputs of consistent length, the sequences were padded or truncated to a maximum length of 50 tokens, allowing the models to process all reviews in a fixed format.

For word representation, the pre-trained GloVe embeddings described in the previous section were used as the basis for the embedding layer. The embedding matrix was aligned with the custom vocabulary so that each token index corresponds to its respective 100-dimensional vector. Words not found in the pre-trained embedding set were assigned small random vectors so that they could still be learned during training. The embedding layer was configured with trainable parameters, allowing the neural network to fine-tune the vectors and adapt them to domain-specific language commonly found in gaming reviews.

To evaluate different approaches to modelling textual sentiment, three neural network architectures were implemented and compared: a Simple Recurrent Neural Network (RNN), a Long Short-Term Memory (LSTM) network, and a hybrid Convolutional Neural Network with Bidirectional LSTM (Conv1D + BiLSTM) model.

The Simple RNN model serves as a baseline architecture. It consists of an embedding layer followed by a SpatialDropout1D layer, a SimpleRNN layer with 128 units, a fully connected dense layer with ReLU activation, and a final output layer with a sigmoid activation function. The SpatialDropout1D layer randomly disables entire embedding dimensions during training, helping to regularise the embedding representation and reduce overfitting. The SimpleRNN layer processes the text sequentially and learns patterns based on word order. However, traditional RNNs can struggle to retain information from earlier parts of a sequence due to the vanishing gradient problem.

To address this limitation, a second architecture replaces the SimpleRNN layer with a LSTM layer. LSTMs introduce internal gating mechanisms that regulate the flow of information through the network. These gates allow the model to retain important contextual information across longer sequences, making LSTMs particularly suitable for natural language processing tasks. In this architecture, the LSTM layer is followed by a dense layer with ReLU activation, a Dropout layer for regularisation, and a final sigmoid output layer that predicts the probability of a positive review.

We decided to include a more advanced architecture combining convolutional and recurrent neural network techniques. This model consists of an embedding layer, SpatialDropout1D regularisation, a Conv1D layer with 128 filters and a kernel size of three, a Bidirectional LSTM layer, GlobalMaxPooling1D, and several fully connected layers. The convolutional layer acts as a feature extractor by identifying local n-gram patterns in the text, such as short phrases that strongly suggest sentiment. The

Bidirectional LSTM processes the review sequence in both forward and backward directions, allowing the model to capture contextual information from both preceding and following words in the sequence. This helps the network understand more complex sentiment expressions.

The output of the Bidirectional LSTM is passed to a GlobalMaxPooling1D layer, which extracts the strongest feature signals across the sequence and converts the variable-length sequence representation into a fixed-length feature vector. This representation is then processed by a dense layer with ReLU activation, followed by Batch Normalization, which stabilises and accelerates training by normalising intermediate activations. A Dropout layer is also applied to reduce overfitting by randomly deactivating neurons during training. Finally, a single neuron with a sigmoid activation function produces the probability that the review expresses positive sentiment.

All models were trained using the binary cross-entropy loss function, which is appropriate for binary classification tasks. The Adam optimiser was used due to its efficiency and adaptive learning rate capabilities. Since the dataset is highly imbalanced, with a significantly larger number of positive reviews than negative ones, we decided to implement a class weighting during training to ensure that minority class samples were given appropriate importance.

To improve training stability and prevent overfitting, early stopping was implemented. This technique monitors the validation loss during training and stops the training process if the model's performance stops improving for several epochs. In addition, the final model incorporated a learning rate scheduling strategy using ReduceLROnPlateau, which automatically reduces the learning rate when validation performance plateaus. This allows the model to make finer adjustments to its weights during later training stages.

In order to further improve model performance, several hyperparameters were automatically tuned. A grid search approach was used to explore different combinations of batch size, dropout rate, and learning rate. Specifically, batch sizes of 64, 128, and 256, dropout rates of 0.1, 0.2, and 0.3, and learning rates of 0.001, 0.0005, and 0.0001 were evaluated. Each combination of parameters was tested using the Conv1D + Bidirectional LSTM architecture, which showed the strongest initial performance.

For each hyperparameter configuration, the model was trained for several epochs and evaluated using multiple performance metrics, including accuracy, precision, recall, and macro F1-score. Because the dataset is imbalanced, the macro F1-score was used as the primary metric, as it provides a balanced measure of performance across both classes. The grid search procedure recorded the results for each configuration and selected the parameter combination that achieved the highest macro F1-score.

The final model therefore uses the Conv1D + Bidirectional LSTM architecture with the optimal hyperparameters identified through grid search. This model integrates pre-trained word embeddings, convolutional feature extraction, bidirectional sequence

modelling, and regularisation techniques such as dropout and batch normalization. Through this combination of architectural design and hyperparameter optimisation, the model is able to effectively capture both local sentiment patterns and broader contextual relationships within the review text.

1.2.3 Evaluation and Insights

To assess the effectiveness of the final sentiment classification model, several evaluation metrics and visual diagnostics were used. These include learning curves, confusion matrices, classification reports, and threshold analysis, all generated directly by the notebook in the evaluation section. Together, these metrics provide a comprehensive understanding of the model's predictive performance, training behaviour, and robustness when applied to an imbalanced sentiment classification task.

The notebook first evaluates the training dynamics of the model through training and validation learning curves, which illustrate how performance evolves across epochs.

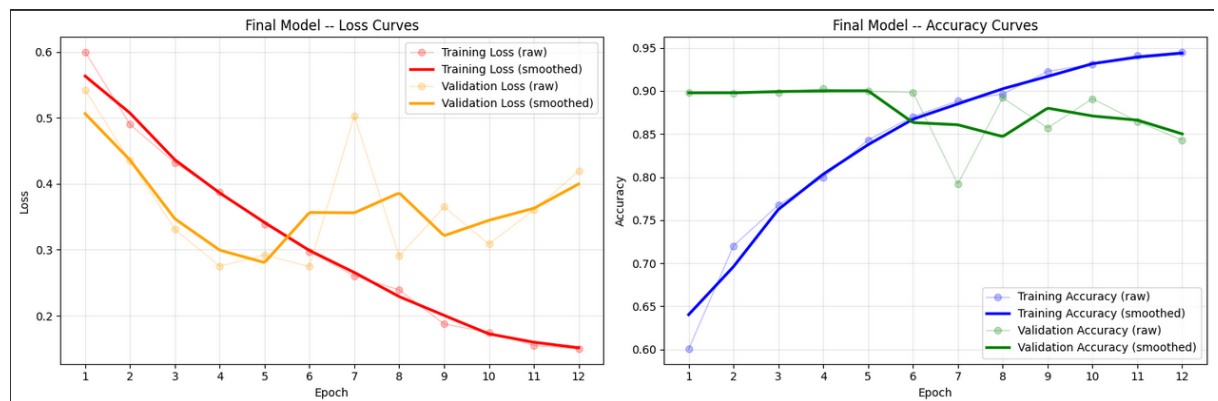


Figure 1: Training and validation loss and accuracy curves

Figure 1 shows both training and validation loss curves alongside accuracy curves. These plots allow direct observation of how the model learns during training. The loss curves demonstrate whether the model is successfully minimising prediction error over time, while the accuracy curves provide insight into how well the model performs on both training and validation datasets.

To make the trends easier to interpret, the curves are smoothed while still preserving the raw epoch values. The plots indicate that the model converges steadily, with training and validation metrics following similar trajectories. This suggests that the combination of dropout regularisation, batch normalisation, and early stopping helps prevent severe overfitting while allowing the model to learn meaningful patterns from the data.

Confusion Matrix Analysis

To better understand how the model performs across classes, a confusion matrix was generated for the final predictions.

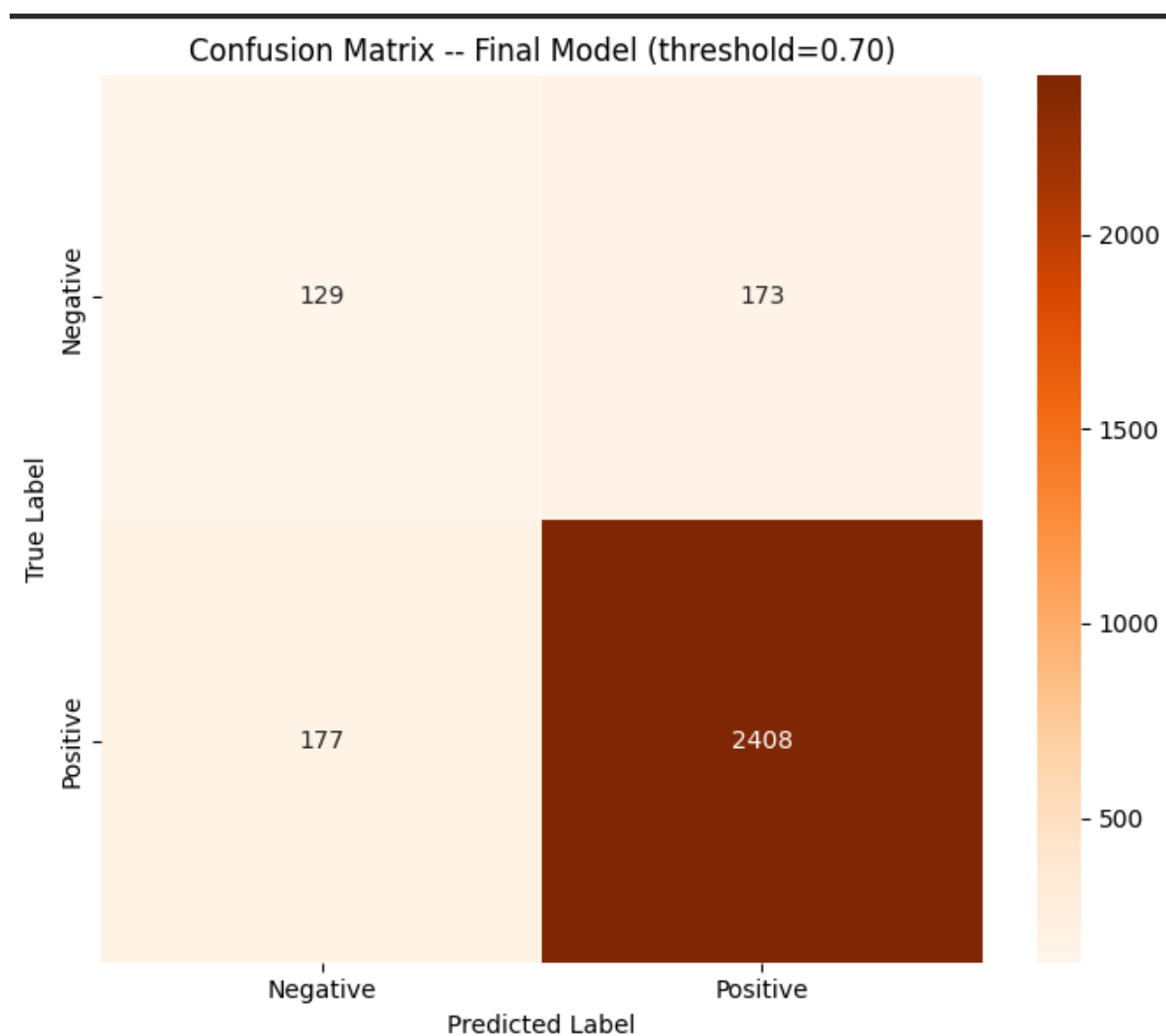


Figure 3: Confusion matrix

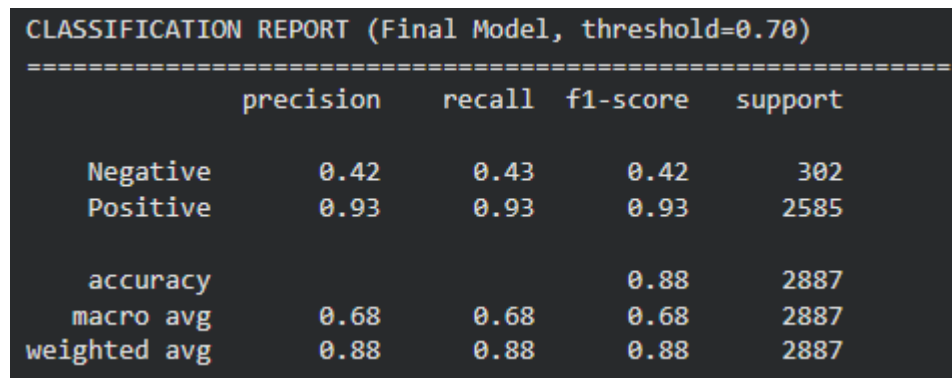
The confusion matrix visualises the number of true negatives, false positives, false negatives, and true positives predicted by the model. This representation provides a clear view of how well the model distinguishes between positive and negative reviews. Because the dataset is highly imbalanced (approximately 89% positive and 11% negative reviews), examining the confusion matrix is particularly important to ensure that the model does not simply predict the majority class.

The heatmap shows that the model correctly identifies a large number of positive reviews while also maintaining reasonable performance on the minority negative class. Although some misclassifications remain, particularly among very short or ambiguous

reviews, the confusion matrix demonstrates that the model is capable of learning meaningful sentiment patterns despite the imbalance.

Classification Metrics

In addition to the confusion matrix, the final model was evaluated using several standard classification metrics.



CLASSIFICATION REPORT (Final Model, threshold=0.70)				
	precision	recall	f1-score	support
Negative	0.42	0.43	0.42	302
Positive	0.93	0.93	0.93	2585
accuracy			0.88	2887
macro avg	0.68	0.68	0.68	2887
weighted avg	0.88	0.88	0.88	2887

Figure 4: Final classification report summary

The classification report includes accuracy, precision, recall, and F1-score for each class. While overall accuracy provides a useful measure of general predictive performance, it can be misleading in imbalanced datasets. For this reason, the macro F1-score is emphasised in the evaluation. Macro F1 treats each class equally by averaging F1-scores across classes, making it a more reliable indicator of balanced performance when class distributions are uneven. By analysing these metrics together, it becomes clear how well the model balances correctly identifying positive reviews while still detecting negative ones.

Decision Threshold Optimisation

Although the default probability threshold for binary classification is typically 0.5, the notebook performs additional analysis to confirm whether this value is optimal for the dataset.

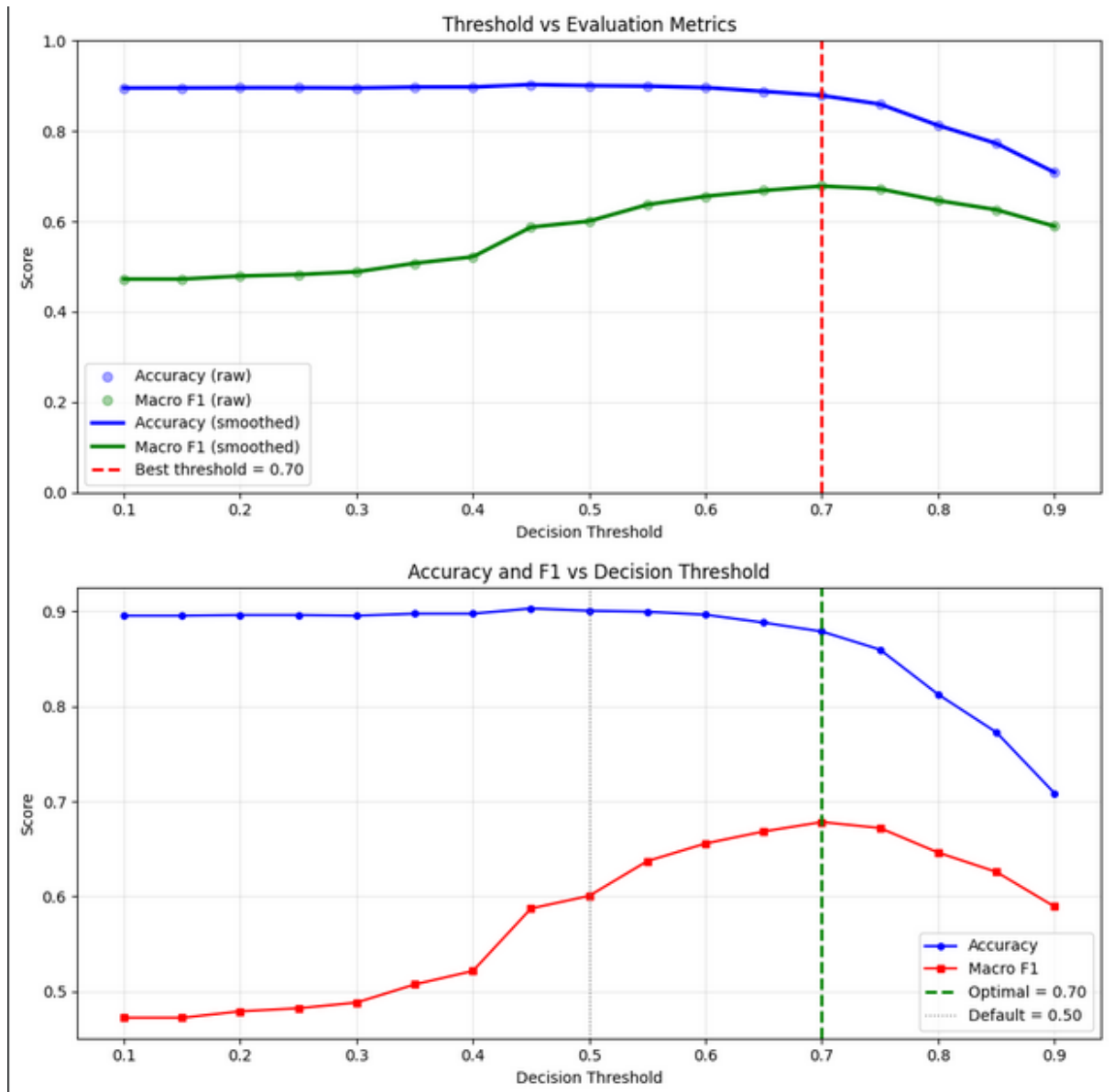


Figure 2: Decision threshold sweep showing accuracy and macro F1 across thresholds

To perform this analysis, the decision threshold was systematically varied from 0.10 to 0.90, and both accuracy and macro F1-score were computed for each threshold value. This sweep allows the model's performance to be examined across different operating points. The results show that the 0.5 threshold provides the best balance between precision and recall, resulting in the highest macro F1-score. This confirms that the standard threshold remains appropriate for the final model configuration.

Strengths of the Approach

The evaluation highlights several strengths of the implemented sentiment classification system. First, the custom preprocessing pipeline provides full control over the data cleaning and tokenisation process, ensuring that noisy or irrelevant content is removed before training. This improves the quality of the textual input provided to the model.

Second, the comparison of multiple neural architectures allows the most effective model design to be identified. Among the tested models, the Conv1D + Bidirectional LSTM architecture demonstrates superior performance by combining local pattern extraction with contextual sequence modelling. Third, the use of grid search hyperparameter tuning and class weighting helps mitigate the effects of the strong class imbalance in the dataset. By adjusting batch size, learning rate, and dropout rate, the training process identifies parameter combinations that maximise macro F1-score.

Another advantage is the use of trainable GloVe embeddings, which allow the model to refine pre-trained word representations and adapt them to domain-specific language commonly found in gaming reviews. Finally, the evaluation process itself is comprehensive. By combining learning curves, confusion matrices, classification metrics, and threshold analysis, the notebook provides clear visual confirmation of model behaviour, as illustrated in Figures 1–4.

Limitations

Despite the strong performance of the model, several limitations remain. Sentiment classification models based on recurrent networks may struggle to interpret sarcasm, irony, or complex linguistic structures, which are common in informal online reviews. For example, phrases that appear positive in isolation may actually convey negative sentiment depending on context. In addition, the dataset contains very short reviews and potentially noisy labels, which can limit the maximum achievable performance. Extremely brief comments may lack sufficient context for accurate sentiment prediction, even when advanced neural architectures are used.

Future Improvements

Future work could explore the use of transformer-based models, such as Bidirectional Encoder Representations from Transformers (BERT). Transformer architectures use self-attention mechanisms to capture relationships between words across an entire sentence simultaneously. Pre-trained transformer models also incorporate rich contextual embeddings learned from very large corpora, which could significantly improve performance on nuanced sentiment expressions and sarcasm. Integrating such models could further enhance the system's ability to understand complex review language and achieve higher classification accuracy.

Summary

Overall, the evaluation demonstrates that the developed sentiment classification system performs effectively on the CS:GO Steam reviews dataset. The notebook follows several industry standard practices for evaluating machine learning models, including

the use of learning curves (Figure 1), confusion matrices (Figure 3), classification metrics (Figure 4), and threshold optimisation analysis (Figure 2). These evaluation techniques provide quantitative and visual insight into model performance, ensuring that the chosen architecture and hyperparameters are well justified. Together, they confirm that the final Conv1D + Bidirectional LSTM model is capable of capturing meaningful sentiment patterns within the Steam review text while maintaining balanced performance across classes.